Decide & Defend - Piecewise Onto Functions

Joy and Marianna have written two different sets of code to accomplish the same goal of helping a caterer direct people with dietary restrictions to a menu item that works for them. Look at the code below.

Joy's Code:

```
(define (entree diet)
  (cond
    [(string=? diet "gluten-free") salmon]
    [(string=? diet "kosher") salmon]
    [(string=? diet "lactose-int") salmon]
    [(string=? diet "nut allergy") lasagna]
    [(string=? diet "vegan") stir-fry]
    [(string=? diet "vegetarian") stir-fry]
    [else (text "unknown diet" 20 "red")])
```

Marianna's Code:

```
(define (entree diet)
  (cond
    [(or (or (string=? diet "gluten-free") (string=? diet "kosher")) (string=? diet "lactose-int"))
salmon]
    [(or (string=? diet "vegan") (string=? diet "vegetarian")) stir-fry]
    [else (text "unknown diet" 20 "red")])
```

Whose method do you like better? Why?

_____

_____

_____

_____

Mood Generator - Explore

1) Open the Mood Generator Starter File, and read through the code you find there. This code contains new programming that you haven't seen yet! Take a moment to list everything you Notice, and then everything you Wonder…

Notice	Wonder

2) Add another line of code to the definition for a "mad" mood that produces the *same* emoji as as an "angry" mood.

3) Add **another example** for "laughing", using the appropriate emoji. *(To bring up the emojis on your computer, type* `Cmd-Ctrl-Space` *on a Mac, or* `Windows-Period` *on Windows 10.)*

4) Come up with some new moods, and add them to the code. Make sure you include examples!

5) In your own words, describe how *piecewise functions* work in this programming environment.

_____

_____

_____

_____

Word Problem: Mood Generator

NOTE: This file uses emojis. Even though emojis look like images, they are actually characters in a string! They can be accessed from your keyboard, just like any other character.

Directions: They say a picture is worth a thousand words. Write a function `mood` that translates moods into emojis so that we can "see" what someone is feeling.

Contract and Purpose Statement

Every contract has three parts...

; *mood* : *String* -> *String*

function name *Domain* *Range*

; *Consumes a mood and produces the emoji for that mood.*

what does the function do?

Examples

Write some examples, then circle and label what changes...

(EXAMPLE (mood "happy") "😀")

function name *input(s)* *what the function produces*

(EXAMPLE (mood "sad") "😢")

function name *input(s)* *what the function produces*

(EXAMPLE (mood "angry") "😡")

function name *input(s)* *what the function produces*

(EXAMPLE (mood "sick") "🤮")

function name *input(s)* *what the function produces*

Definition

Write the definition, giving variable names to all your input values...

```
(define (                                   )
                                   )
  (cond
    [                                   ]
    [                                   ]
    [                                   ]
    [                                   ]
    [                                   ] ))
```

function name *variable(s)*

Welcome to Alice's Restaurant!

Alice has hired you to improve some code used at the restaurant. The code we'll be improving on is shown below.

Read through the code line-by-line with your partner before writing down your observations in the tables below.

```
; cost :: String -> Number
; given a item, produce the cost of that item
```

```
(define (cost item)
  (cond
    [(string=? item "onion rings") 3.5]
    [(string=? item "fried tofu") 5.25]
    [(string=? item "pie") 2.25]
    [else "Sorry, that's not on the menu!"])
```

1) **I notice...**	2) **I wonder...**
3) **Familiar things I see in the code**	4) **Unfamiliar things I see in the code**

Alice's Restaurant - Explore

Alice's code has some new elements we haven't seen before, so let's experiment a bit to figure out how it works! **Open the [Alice's Restaurant Starter File](), click "Run", and try using the `cost` function in the Interactions window.**

1) What does `(cost "hamburger")` evaluate to? _____

2) What does `(cost "pie")` evaluate to? _____

3) What if you ask for `(cost "fries")`? _____

4) Explain what the function is doing in your own words. _____

_____

_____

5) What is the function's name? Domain? _____

6) What is the name of its variable? _____

7) Alice says onion rings have gone up to $3.75. Change the `cost` function to reflect this.

8) Try adding menu items of your own. What's your favorite?

9) For an unknown food item, the function produces the String `"That's not on the menu!"` Is this a problem? Why or why not?

_____

_____

10) Suppose Alice wants to calculate the price of a hamburger, *including a 5% sales tax*. Draw a Circle of Evaluation for the expression below.

Word Problem: Alice's Restaurant

Directions: Alice's Restaurant has hired you as a programmer. They offer the following menu items: hamburger ($6.00), onion rings ($3.50), fried tofu ($5.25) and pie ($2.25). Write a function called `Alice's Restaurant` which takes in the name of a menu item and outputs the price of that item.

Contract and Purpose Statement

Every contract has three parts…

; _____ : _____ -> _____
function name *Domain* *Range*

; _____
what does the function do?

Examples

Write some examples, then circle and label what changes…

(EXAMPLE (_____) _____)
function name *input(s)* *what the function produces*

(EXAMPLE (_____) _____)
function name *input(s)* *what the function produces*

(EXAMPLE (_____) _____)
function name *input(s)* *what the function produces*

(EXAMPLE (_____) _____)
function name *input(s)* *what the function produces*

Definition

Write the definition, giving variable names to all your input values…

(define (_____)
function name *variable(s)*

(*cond* _____

[_____ _____]

[_____ _____]

[_____ _____]

[_____ _____]

[_____ _____]))